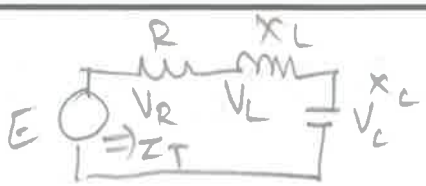
Applying VDR,



$$V_L = E\frac{X_L}{Z_T} = E\frac{X_L}{R} \text{ (at resonance)}$$

$$\boxed{V_{L_s} = Q_s E}$$

Similarly,

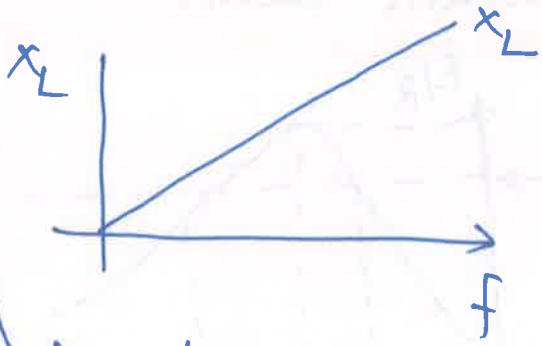
$$V_C = E\frac{X_C}{Z_T} = E\frac{X_C}{R} \text{ (at resonance)}$$

$$\boxed{V_{C_s} = Q_s E}$$

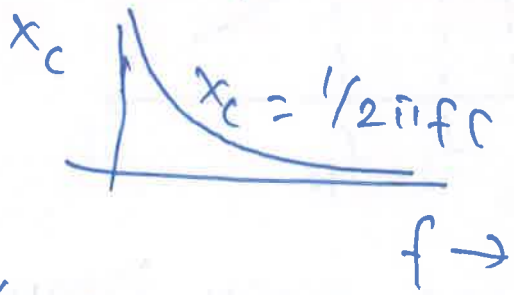
Effect of frequency on Z_T:

$$Z_T = R + j(X_L - X_C)$$

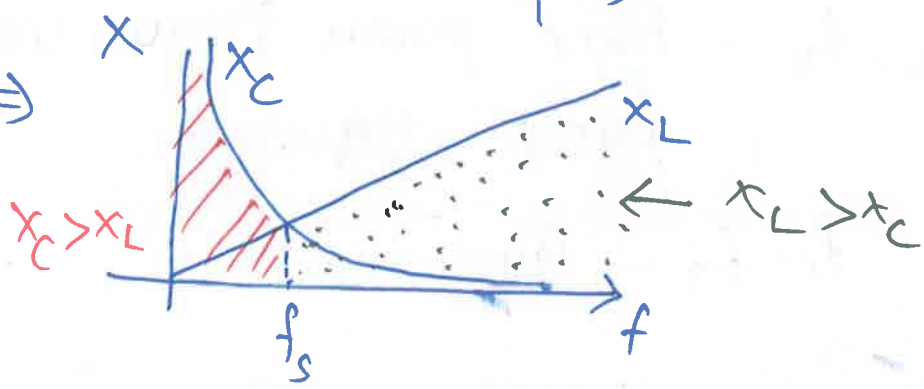
$$X_L = 2\pi f L$$



$$X_C = \frac{1}{2\pi f C}$$



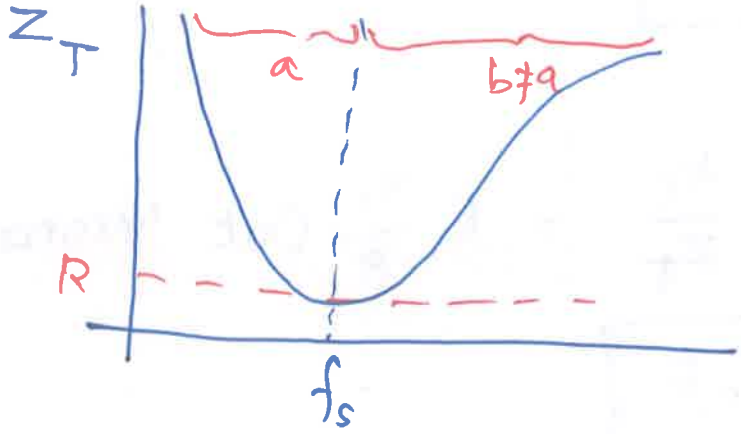
$$X = X_L - X_C \Rightarrow$$



$f < f_s \rightarrow X_C > X_L \rightarrow$ n/w capacitive $\rightarrow$ I leads E

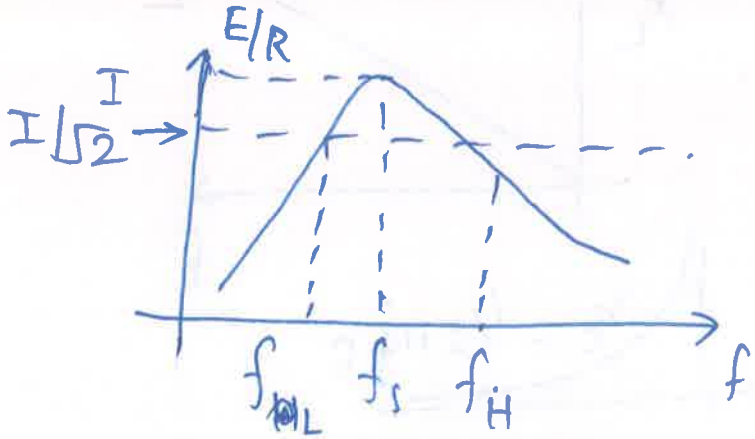
$f > f_s \rightarrow X_L > X_C \rightarrow$ n/w ~~cap~~ inductive $\rightarrow$ E leads I

$f = f_s \rightarrow$ ~~Ntw~~ $X_L = X_C \rightarrow$ n/w resistive $\rightarrow$ E, I in same phase.



Selectivity:

Plotting I Vs frequency, where $I = E/Z_T$, E = Constant value.



f_L, f_H - half power frequencies, cut off freq. band frequency.

$f_1 - f_2$ - Bandwidth.

Half power frequencies:

Those frequencies at which the power delivered is one-half that delivered at the resonant frequency

$$P_{HF} = \frac{1}{2} P_{max}$$

Proof:

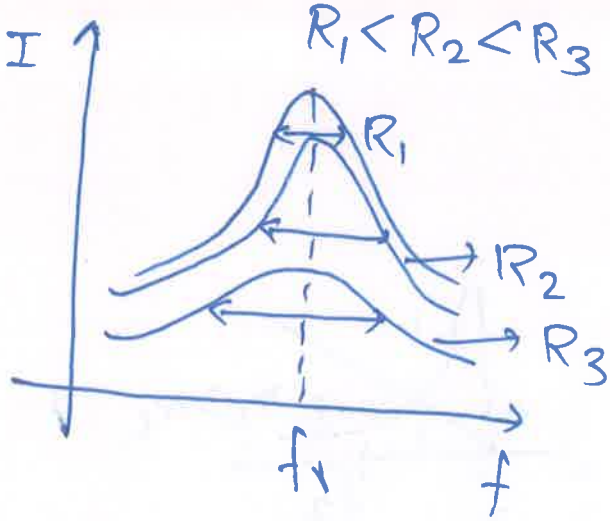
$$P_{max} = I_{max}^2 R \quad [\text{at resonance } f_r].$$

$$P_{HPF} = (0.707\, I_{max})^2 R = 0.5\, I_{max}^2 R = \frac{1}{2} P_{max}.$$

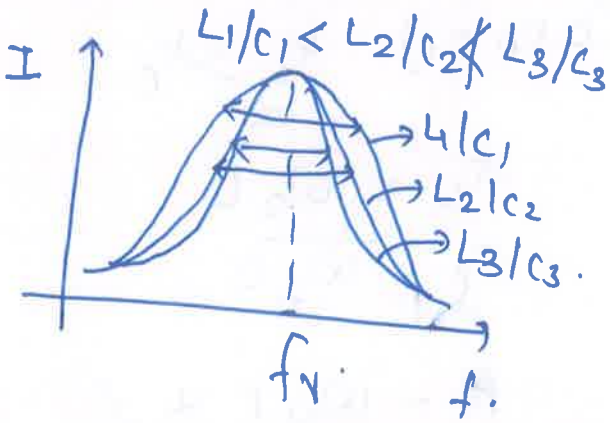
Selectivity curve:-

→ derived from the fact that one must be selective of the frequency.

→ Smaller BW → higher is selectivity.



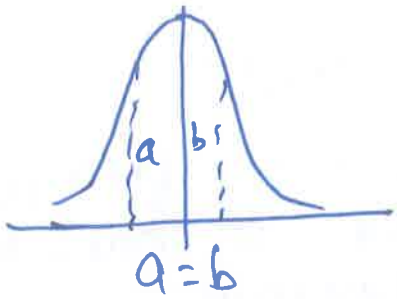
(L, C - fixed)



R - fixed.

Smaller (R) ⇒ higher selectivity and vicaversa. ⇒ $Q = \frac{X_L}{R}$ (larger)

If $Q \geq 10$



A drop in current by 0.707 $\Rightarrow$ ↑ the Impedance by 0.707 for a constant E.

At half power:-

$$0.707\, Z_T \equiv 0.707\, R = \sqrt{2}\, R$$

$$\therefore \quad Z_T = \sqrt{R^2 + (x_L - x_C)^2}$$

$$\sqrt{2}\, R = \sqrt{R^2 + (x_L - x_C)^2}$$

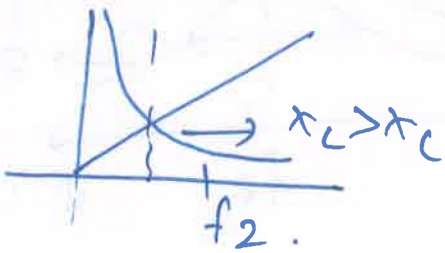
$$2R^2 = R^2 + (x_L - x_C)^2$$

$$R^2 = (x_L - x_C)^2$$

$$R = x_L - x_C$$

$$\Rightarrow \quad R - x_L + x_C = 0$$

Case (i) $\underline{x_L > x_C}$:



$$x_L = \omega_2 L$$

$$x_C = \frac{1}{\omega_2 C}$$

$$R - \omega_2 L + \frac{1}{\omega_2 C} = 0$$

$$R\omega_2 - \omega_2^2 L + \frac{1}{C} = 0.$$

$$\underset{a}{\omega_2^2} - \underset{b}{\frac{R}{L}}\,\omega_2 + \underset{c}{\frac{1}{LC}} = 0.$$

$$\omega_2 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{R}{2L} \pm \frac{1}{2}\sqrt{\left(\frac{R}{L}\right)^2 - 4\left(\frac{1}{LC}\right)}$$

$$f_2 = \frac{1}{2\pi}\left[\frac{R}{2L} + \frac{1}{2}\sqrt{\left(\frac{R}{L}\right)^2 + \frac{4}{LC}}\right.$$

– – dropped because $\frac{R}{2L}$ 2nd term is always $> R/2L$

Similarly for $X_C > X_L \Rightarrow f_1$

$$f_1 = \frac{1}{2\pi}\left[-\frac{R}{2L} + \frac{1}{2}\sqrt{\left(\frac{R}{L}\right)^2 + \frac{4}{LC}}\right]$$

$$BW = f_2 - f_1 = \frac{1}{2\pi}\left[\frac{R}{2L} + \frac{R}{2L}\right] = \frac{R}{2\pi L}$$

$$BW = \frac{1}{2\pi}\left[\frac{R}{L}\right] = \left(\frac{f_s}{\omega_s}\right)\left(\frac{\omega_s}{Q_s}\right) \leftarrow Q = \frac{R}{\omega L}$$

$\omega = 2\pi f$

$$\boxed{BW = \frac{f_s}{Q_s}}$$

$$f_2 - f_1 = BW.$$

$$\Rightarrow (f_2 - f_1)/f_s = BW/f_s = 1/Q_s.$$

$\frac{f_2 - f_1}{f_s}$ => fractional Bandwidth.

$$\boxed{f_s = \sqrt{f_1 f_2}.}$$

Frequencies at which V_c and V_L are maximum

$V_c = I x_c$ in series resonance circuit.

where, $I = \frac{V}{Z_T} = \frac{V}{\sqrt{R^2 + (x_L - x_c)^2}}$

$$V_c = \left[\frac{V}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega c}\right)^2}}\right] \cdot \frac{1}{\omega c}$$

$$V_c^2 = \frac{V^2}{(\omega c)^2 \left[R^2 + \left(\omega L - \frac{1}{\omega c}\right)^2\right.}$$

$$= \frac{V^2}{\omega^2 c^2 R^2 + \omega^2 c^2 \left(\frac{\omega^2 L c - 1}{\omega c}\right)^2}$$

$$= \frac{V^2}{\omega^2 c^2 R^2 + (\omega^2 L c - 1)^2}$$

$$\frac{dV_c^2}{d\omega} = 0 \Rightarrow \frac{V^2\left(2\omega c^2 R^2 + 2(\omega^2 L c - 1)(2\omega L c)\right)}{\left[\omega^2 c^2 R^2 + (\omega^2 L c - 1)^2\right]^2}$$

$$2\omega C^2R^2 + 2(\omega^2LC-1)(2\omega LC) = 0$$

$$CR^2 + 2(\omega^2LC-1)L = 0.$$

$$2\omega^2L^2C = 2L - R^2C$$

$$\omega^2 = \frac{L}{L^2C} - \frac{R^2C}{2L^2C}$$

$$\omega = \sqrt{\frac{1}{LC} - \frac{R^2}{2L^2}}$$

$$\boxed{f_c = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{2L^2}}.}$$

$$V_L = I x_L$$

$$V_L = \frac{V(\omega L)}{\sqrt{R^2 + (\omega L - 1/\omega C)^2}}$$

$$V_L^2 = \frac{V^2\omega^2L^2}{R^2 + (\omega L - \frac{1}{\omega C})^2} = \frac{V^2\omega^4L^2C^2}{\omega^2C^2R^2 + (\omega^2LC-1)^2}$$

$$\frac{dV_L}{d\omega} = 0 \Rightarrow \{V\,dv/dx - u\,dV/dx\}/V^2$$

A1. Q2

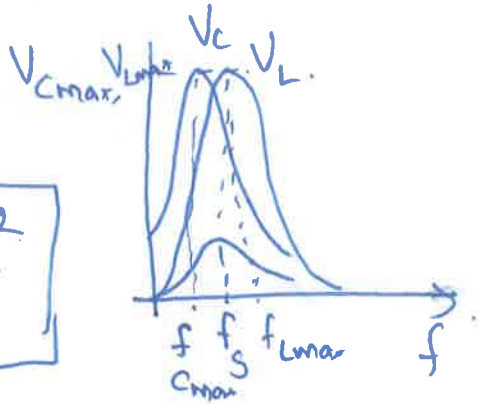
$$\left\{ -V^2\omega^2L^2\frac{\left[2\omega C^2R^2 + 2(\omega^2LC-1)^2(2\omega LC)\right]}{\cancel{(\omega^2C^2R^2 + (\omega^2LC-1)^2)^2}} + \left[(\omega^2C^2R^2) + (\omega^2LC-1)^2\right]\left[4\omega^3V^2L^2C^2\right]\right\}/V^2$$

$$2\omega^2 LC - \omega^2 C^2 R^2 - 2 = 0$$

$$\omega^2 [2LC - C^2R^2] = 2$$

$$\omega^2 = \frac{2}{2LC - C^2R^2}$$

$$\boxed{f_L = \frac{1}{2\pi}\sqrt{LC - \frac{R^2C^2}{2}}}$$



Eg: 101 A Series RLC has $R = 0.2\Omega$, $L = 100mH$ and $C = 50\mu F$. find f_r and I_{in} at 24V.

$$f_r = \frac{1}{2\pi\sqrt{LC}} = 71.21\,Hz.$$

$$I = \frac{V}{R} = \frac{24}{0.2} = 120A.$$

Eg: 102 $R = 10\Omega$, $L = 25mH$, $C = 100\mu F$?

Find Q Factor?

$$f_r = \frac{1}{2\pi\sqrt{LC}} = 100.71\,Hz.$$

$$Q = \frac{\omega L}{R} = \frac{2\pi(100.71) \times 25 \times 10^{-3}}{10} = 1.58$$

Eg: 103 A 50μF C, when connected in series with a coil having 40Ω resistance, resonates at 1000 Hz. Find the inductance of the coil. Find I if E = 100V. Calculate V_C at resonance; V_{Coil} at resonance.

$C = 50\mu F$. $R = 40$, $L = ?$

$f_r = 1000\ Hz$.

$$1000 = \frac{1}{2\pi\sqrt{L \times 50\times10^{-6}}}$$

$L = 0.5\ mH$.

$$I = \frac{E}{R} = \frac{100}{40} = 2.5\ A.$$

$$V_C = I X_C = 2.5 \times \frac{1}{\cancel{2\pi}\,\omega C} = \frac{2.5}{2\pi \times 1000 \times 50\times10^{-6}}$$

$V_C = 7.96\ V$.

$$V_{coil} = I Z_{coil}$$

$X_L = \omega L = 2\pi \times 1000 \times 0.5 \times 1$

$$= 2.5\left(\sqrt{40^2 + (\cancel{0.5}\ 3.14)^2}\right) = 3.14\ \Omega$$

$= 100.31\ V$.

Eg: 104: Determine the relationship between f_r and f_2 and f_1 in Series resonating circuit.

Refer page 121 $f_r = \sqrt{f_1 f_2}$.

$$f_1 = \frac{1}{2\pi}\left[\frac{-R}{2L} + \frac{1}{2}\sqrt{\left(\frac{R}{L}\right)^2 + \frac{4}{LC}}\right]$$

$$f_2 = \frac{1}{2\pi}\left[\frac{R}{2L} + \frac{1}{2}\sqrt{\left(\frac{R}{L}\right)^2 + \frac{4}{LC}}\right]$$

$$f_1 f_2 = ?$$

Assignment: 1: Q3

A1: Q4a: Obtain the relation $f_1 = f_r - \frac{BW}{2}$

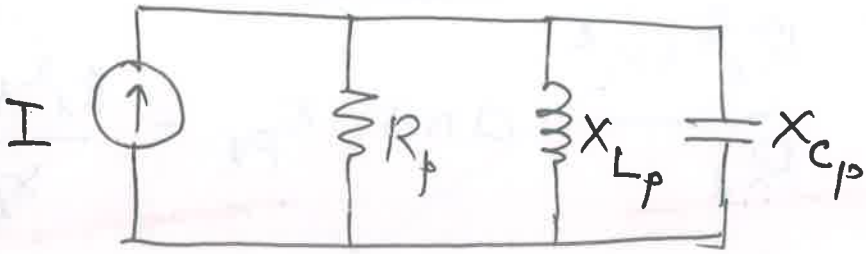
Q4b: $f_2 = f_r + \frac{BW}{2}$

A1: Q5: Rewrite the equation f_c interms of Q

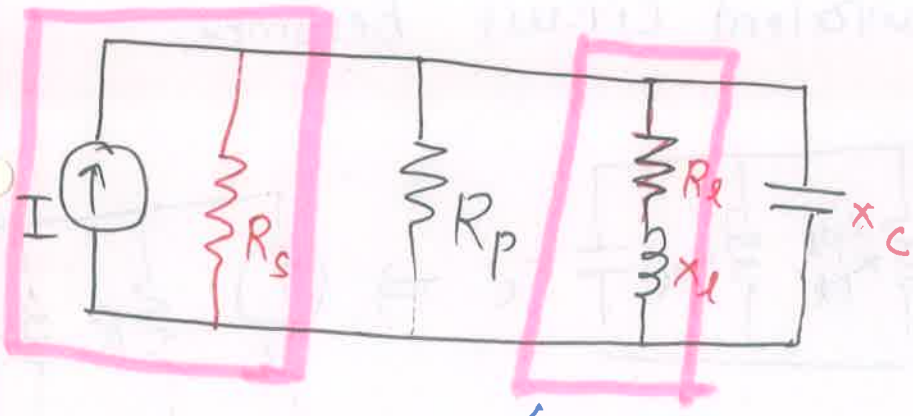
A1: Q6: " " " f_L interms of Q.

Parallel Resonance:

Resonance occuring in parallel combination of R, L and C is called parallel resonance.



But practically, current source has an internal resistance, Inductance has an internal resistance. Therefore, the circuit gets modified to,



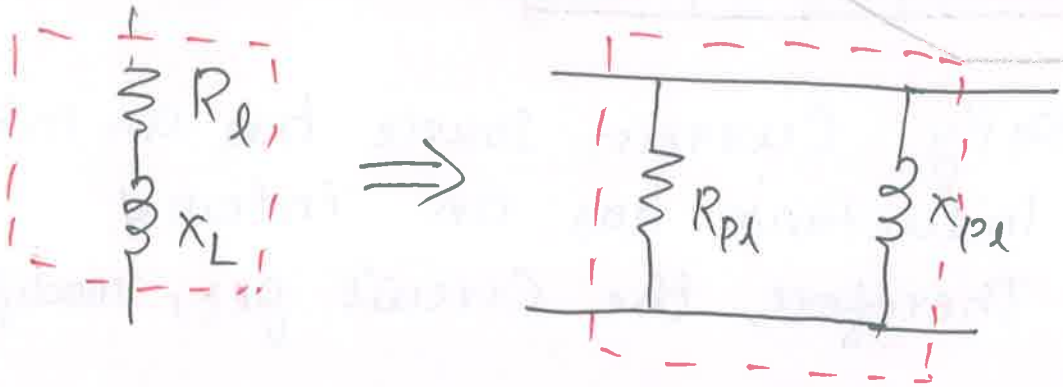
$$Z_{R-L} = R_l + jX_l$$

$$Y_{R-L} = \frac{1}{Z_{RL}} = \frac{1}{R_l + jX_l}$$

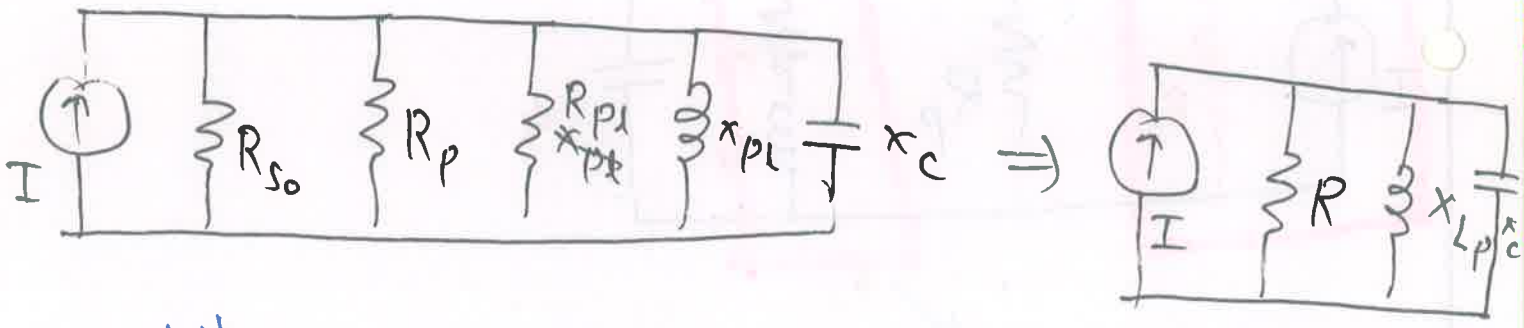
$$Y_{RL} = \frac{R_l - jx_l}{R_l^2 + x_l^2} = \frac{R_l}{R_l^2 + x_l^2} - j\frac{x_l}{R_l^2 + x_l^2}$$

$$\boxed{Y_{RL} = \frac{1}{R_{pl}} + \frac{1}{jx_{pl}}}$$

$$R_{pl} = \frac{R_l^2 + x_L^2}{R_l} \quad \text{and} \quad x_{pl} = \frac{R_l^2 + x_l^2}{x_l}$$



Therefore, the equivalent circuit becomes,



where, $R = R_{so} \parallel R_p \parallel R_{pl}$

Resonant frequency:-

$$Y_T = \frac{1}{R} + \frac{1}{jx_{Lp}} + \frac{1}{-jx_c}$$

$$Y_T = \frac{1}{R} + j\left(\frac{1}{X_C} - \frac{1}{X_{Lp}}\right)$$

For resonance to occur, imaginary term is to be zero.

$$\frac{1}{X_C} = \frac{1}{X_{Lp}}.$$

$$X_C = X_{Lp} = \frac{R_l^2 + X_l^2}{X_l}$$

$$X_C X_L = \frac{1}{C\omega} \cdot \omega L = \frac{L}{C} = R_l^2 + X_l^2$$

$$(2\pi f_p L)^2 = \frac{L}{C} - R_l^2$$

$$\boxed{f_p = \frac{1}{2\pi L}\sqrt{\frac{L}{C} - R_l^2}}$$

Rearranging,

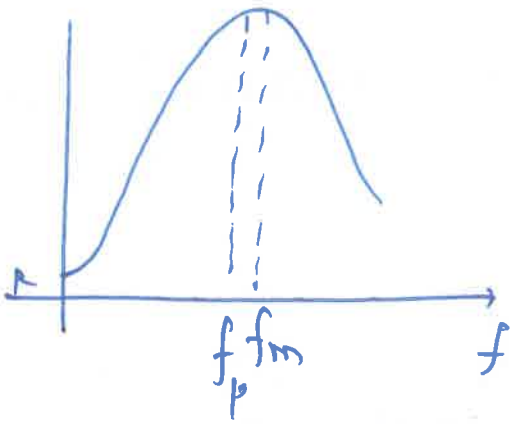
$$f_p = \frac{1}{2\pi L}\sqrt{\frac{1}{(C/L)} - R_l^2}$$

$$= \frac{1}{2\pi L}\sqrt{\frac{1 - R_l^2 (C/L)}{C/L}}$$

$$= \frac{1}{2\pi L\sqrt{C/L}}\sqrt{1 - R_l^2 \cdot \frac{C}{L}}$$

$$f_p = \frac{1}{2\pi\sqrt{LC}}\sqrt{1 - R_l^2 \cdot C/L} = \boxed{f_s\sqrt{1 - R_l^2 \cdot \frac{C}{L}}}$$

Maximum Impedance f_m :-



$f_p \neq f_m$ due to freq. dependence of

$$R_p = \frac{R_l^2 + X_l^2}{R_l}.$$

$$\therefore \frac{dZ_T}{df} = 0 \quad \text{where } Z_T = (R \,||\, X_{Lp} \,||\, X_C)$$

$$f_m = f_s \sqrt{1 - \frac{1}{4}\left(R_l^2 \cdot \frac{C}{L}\right)}$$

$$\boxed{f_s > f_m > f_p.}$$

Half power frequencies in parallel resonance:-

$$Y_T = \frac{1}{R} + j\left(\frac{1}{X_C} - \frac{1}{X_{Lp}}\right)$$

$$|Z| = Z = \frac{1}{\sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2}} = 0.707R$$

Inverting & Squaring both sides,

$$\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L}\right)^2 = \frac{2}{R^2}$$

$$\left(\omega C - \frac{1}{\omega L}\right)^2 = \frac{1}{R^2}$$

$$\omega C - \frac{1}{\omega L} = \pm \frac{1}{R}$$

AI Q7:

Solving the above for positive ω,

$$\omega_1 = -\frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{1}{2RC} + \sqrt{\left(\frac{1}{2RC}\right)^2 + \frac{1}{LC}}$$

$$f_1 = \frac{1}{4\pi C}\left[\frac{-1}{R} + \sqrt{\left(\frac{1}{R^2} + \frac{4C}{L}\right)}\right]$$

$$f_2 = \frac{1}{4\pi C}\left[\frac{1}{R} + \sqrt{\left(\frac{1}{R^2} + \frac{4C}{L}\right)}\right]$$

Effect of $Q \geq 10$:-

let us look now, how various relations gets simplified when $Q \geq 10$.

We know that, $Q = \frac{X_L}{R}$ $\quad \frac{1}{Q} = \frac{1}{10} = 0.1$

$\frac{1}{Q^2} = \frac{1}{100} = 0.01 \ll 1$

Inductive reactance in parallel resonance:-

X_{Lp}:

$$X_{Lp} = \frac{R_l^2 + X_l^2}{X_L} = \frac{R_l^2}{X_L} \cdot \left(\frac{X_L}{X_L}\right) + X_L$$

$$X_{Lp} = \frac{X_L}{Q^2} + X_L$$

$$X_{Lp} \simeq X_L \qquad \left[\frac{1}{Q^2} \ll 1\right].$$

$$X_L \simeq X_C.$$

Resonant frequency:

$$f_p = f_s\sqrt{1 - R_l^2 \cdot \frac{C}{L}}$$

$$R_l^2 \cdot \frac{C}{L} = \frac{1}{\left(\frac{L}{R_l^2 \cdot C}\right)} = \frac{1}{\frac{\omega L}{\omega R_l^2 \cdot C}} = \frac{1}{\frac{\omega L}{R_l^2 \omega C}} = \frac{1}{\frac{X_L X_C}{R_l^2}}.$$

Since $X_L \simeq X_C$ for $Q \geq 10$,

$$\frac{1}{\frac{X_L^2}{R_l^2}} = \frac{1}{Q^2}$$

Q Factor for ||el reso:

$$Q_p = \frac{R}{X_{Lp}} \quad \left\{\because \frac{V^2/X_{Lp}}{V^2/R}\right\}$$

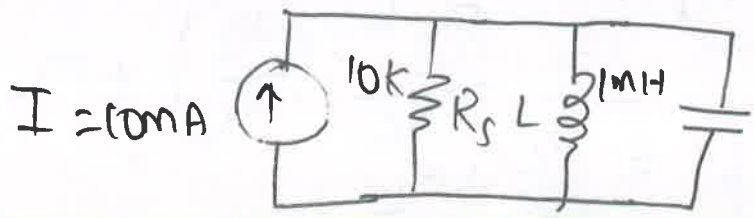
Therefore,

$$f_p = f_s\sqrt{1 - \frac{1}{Q^2}}$$

$$f_p \simeq f_s = \frac{1}{2\pi\sqrt{LC}}$$

problems on parallel resonance:-

Problem: 105:



f_p, Z @ resonance

Q, BW, f_1, f_2

V_C at resonance,

I_L, I_C at resonance.

a] First determine Q to find the nature of the circuit, so as to decide on using approximate expressions.

$$Q = \frac{X_L}{R} \qquad Q_{Coil} = \frac{X_L}{0} = \infty \gg 1$$, so approx. on resonant frequency can be used.

$$f_p = f_s = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{1mH \times 1\mu F}}$$

$$\boxed{f_p = f_s = 5.03\,kHz}$$

b] $Z = R_s \| Z_L \| Z_c = R_s$ at resonance

Z at resonance $= 10k\Omega$.

c] $$Q_p = \cancel{\frac{X_L}{R}} = \frac{R}{X_{Lp}} = \frac{10k\Omega}{2\pi f_p L} = \frac{10k}{2\pi(5.03k)(1mH)}$$

$$Q_p = 316.41$$

$$BW = \frac{f_p}{Q_p} = \frac{5.03\,kHz}{316.41} = 15.90\,Hz.$$

$$f_1 = \frac{1}{4\pi C}\left[\frac{-1}{R} + \sqrt{\frac{1}{R^2} + \frac{4C}{L}}\right] = 5.03\,kHz.$$

$$f_2 = \frac{1}{4\pi C}\left[\frac{1}{R} + \sqrt{\frac{1}{R^2} + \frac{4C}{L}}\right] = 5.04\,kHz.$$

d] $V_C = I X_C = \cancel{(10mA)}$

At resonance

$$V_C = V_R = I R_s = (10mA)(10k)$$

$$V_C\big|_{Reso.} = 100V$$

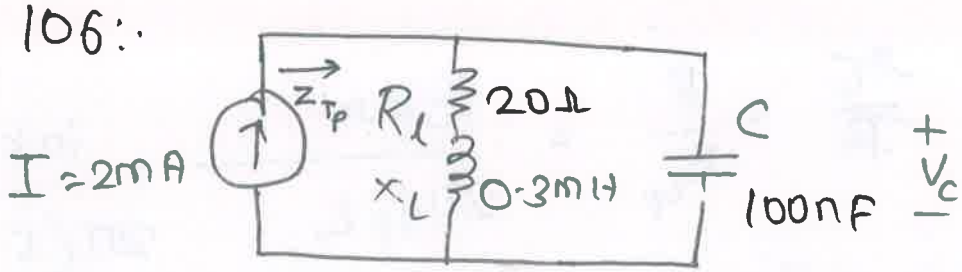
e] $$I_L = \frac{V_L}{2\pi f_p L} = \frac{100}{2\pi(5.03k)(1mH)} = 3.16A.$$

$$I_C = \frac{V_C}{(1/2\pi f_p C)} = \frac{100}{31.6\Omega} = 3.16A$$

or

$$I_C = Q\,I$$

Example 106:



$R_s = \infty$, find f_s, f_m, f_p

Z_{Tm}, V_C at f_m

Q_p, BW = ?, Q_p, Z_{Tp}, V_C, BW, if $Q \geq 10$.

$$f_s = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(0.3m)(100nF)}} = 29.06\,kHz$$

$$f_m = f_s\sqrt{1-\frac{1}{4}\left(R_l^2\cdot\frac{C}{L}\right)} = (29.06k)\sqrt{1-\frac{1}{4}\left(\frac{20^2\cdot 100n}{0.3m}\right)}$$

$$f_m = 25.58\,kHz$$

$$f_p = f_s\sqrt{1-R_l^2\cdot\frac{C}{L}} = 27.06\,kHz$$

$f_s > f_p > f_m$ $\Rightarrow$ will result in a low Q n/w.

$$\left.\begin{aligned} f_p - f_m &\simeq 0.5\,kHz \\ f_s - f_p &\simeq 2\,kHz \end{aligned}\right\}$$

Z_{Tm}:

$$Z_{Tm} = (R_l + jX_l)\,||\,-jX_c$$

$$X_L\Big|_{f_m} = 2\pi f_m L = 2\pi(25.58k)(0.3m) = 53.87\,\Omega$$

$$X_c\Big|_{f_m} = \frac{1}{2\pi f_m C} = 55.69\,\Omega.$$

$$R_l + jX_l = 20 + j53.87\,\Omega = 57.46\angle 69.63^\circ$$

$$Z_{Tm} = \frac{(57.46\angle 69.63)(55.69\angle -90^\circ)}{20 + j53.87 - j55.69\,\Omega}$$

$$= 157.34\angle -15.17^\circ.$$

$$V_c\Big|_{f_m} = I Z_{T_m} = (2mA)(159.34) = 318.68\,mV.$$

Q_p:

$$Q_p = \frac{R_{op}}{X_{Lp}} \quad \cancel{\text{with}} \quad \cancel{R = R_s \| R_p} \nearrow \infty$$

$$Q_l = \frac{X_L}{R_l} = \frac{2\pi(27.06k)(0.3m)}{20\Omega} = \frac{51}{20} = 2.55$$

low Q.

BW:

$$BW = \frac{f_p}{Q_p} = \frac{27.06k}{2.55} = 10.61\,kHz.$$

For $Q \geq 10 \Rightarrow f_m = f_p = f_s = 29.06\,kHz.$

$Q_p = 2.74 \uparrow$ from (2.55)

$$Z_{T_p} = Q_l^2 \cdot R_l = (2.74)^2 20 = 150.15\angle 0^\circ$$

$\downarrow$ from $(159.34\angle -15.17)$

$$V_{cmax} = I Z_{T_p} = (2m)(150.15) = 300.33\,mV.$$

$$BW = \frac{f_p}{Q_p} = \frac{29.06k}{2.74} = 10.61\,kHz.$$